

Ballistic Galvanometer-Figure of Merit

Aim:-

1. To determine the figure of merit k for current of the given ballistic galvanometer.
2. To measure a small current using BG and verify with ammeter

Apparatus:

Ballistic galvanometer, 2 volt accumulator, two high range resistance boxes, a low range resistance box, a commutator, lamp and scale arrangement, Micro ammeter and a plug key.

Principle:

Figure of merit of a mirror galvanometer is defined as the current in ampere required to produce a deflection of one millimetre on a scale placed normal to the beam of light at a distance of one metre from the galvanometer mirror.

Let two resistance boxes r and R be connected in series with an accumulator of e.m.f. E volts .

$$\text{Current in the circuit} = \frac{E}{r+R}$$

$$p.d \text{ available across the terminals of } r = \frac{E}{r+R} r$$

If this p.d is applied to a galvanometer of resistance G ,

$$\text{current flowing through the galvanometer} = \frac{Er}{(r+R)G}$$

The current through the galvanometer $I_g = k d$

Let d be the steady deflection produced in the galvanometer, then

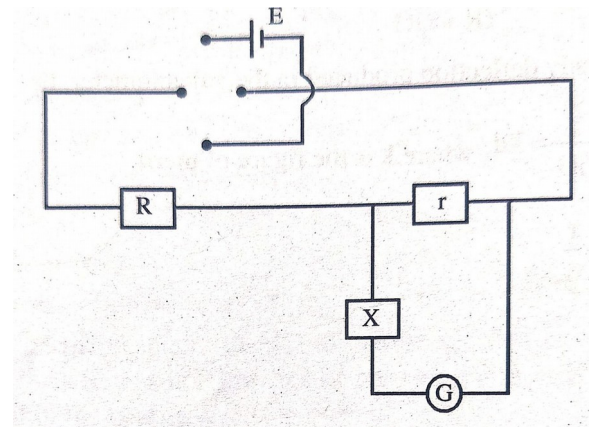
$$\frac{Er}{(r+R)G} = kd$$

$$k = \frac{E}{(r+R)G} \frac{r}{d}$$

To find G :

The value of G , i.e., the resistance of the mirror galvanometer is determined by half deflection method.

$$\text{When } X=0, \text{ we have, } k = \frac{E}{(r+R)G} \frac{1}{d} \frac{r}{d} \dots\dots(1)$$



Introduce a suitable resistance in the box X such that the deflection in the galvanometer is $\frac{d}{2}$, then

$$k = \frac{E}{(r+R)} \frac{1}{G+X} \frac{r}{d/2} \dots\dots\dots (2)$$

Comparing equations (1) and (2), we get

$$\frac{E}{(r+R)} \frac{1}{G} \frac{r}{d} = \frac{E}{(r+R)} \frac{1}{(G+X)} \frac{2r}{d}$$

$$\text{or } \frac{1}{G} = \frac{2}{(G+X)}$$

$$G + X = 2G$$

$$X = G$$

Thus the resistance G of the galvanometer is equal to the resistance X. It is included in series with the galvanometer.

Procedure:

The accumulator E is connected in series with a low resistance box r and a high resistance box R through a commutator. A high range resistance box X and the mirror galvanometer are connected across the terminals of r.

The lamp and scale arrangement is kept in front of the galvanometer such that the image of the vertical cross wire within the spot of light (obtained due to reflection of light from the galvanometer) on the translucent scale is very clear. The spot of light is adjusted to be at the centre of the scale.

A small resistance $r = 1$ ohm is included in r, no resistance is included in X and a high resistance $R = 10000$ ohms is included in R. The circuit is closed and the steady deflection d in the scale is noted. Then resistances are carefully included in X so that the deflection is reduced exactly to half the original value. This gives the resistance of the galvanometer. Then the plug keys in X are replaced so that $X = 0$.

To find the Galvanometer Resistance G:

Using the commutator the current is reversed to get the deflection in opposite direction. The steady deflection is also noted. The mean value of steady deflection is found out. As before resistances are introduced in X and the resistance of the galvanometer is again found out. The mean value of the two resistances be G. This gives the resistance of the galvanometer.

To find the Figure of Merit of Galvanometer k:

The experiment is repeated for different values of r, calculate the figure of merit of the galvanometer using the equation

$$k = \frac{E}{(r+R)} \frac{1}{G} \frac{r}{d}$$

To measure a small Current:

Replace the Galvanometer (G) with a micro-ammeter and adjust the commutator to obtain a positive current. Record the value of this current as I_0 , along with the corresponding deflection d of the Ballistic Galvanometer. Finally, compare the current measured by the BG against the reading from the micro-ammeter.

Observations:

Distance between the galvanometer mirror and the scale ,D= 1 m

E.M.F of the cell,E = 2 Volts

Resistance R=..... Ω

Resistance X to get half deflection:

Left = . . .

Right = . . .

Mean

G=..... Ω

Figure of Merit Determination:

Trial No	Resistance r in Ω	Deflection of the spot d in cm (When X=0)			Figure of merit $k = \frac{E}{(r+R)} \frac{1}{G} \frac{r}{d}$ in A/cm
		Left	Right	Mean d	
1	1				
2	2				
3	3				
4	4				
5	5				

Mean Value of k=

Ammeter Reading vs BG: Comparison

Trial No	Resistance r in Ω	Deflection of the spot d in cm (When $X=0$)			Current measured by BG, $I_g = k d$ (μA)	Ammeter Reading I_g (μA)
		Left	Right	Mean d		
1						
2						
3						
4						
5						

Result:

1. The figure of merit of the mirror galvanometer = ----- A/cm.
2. The values of current calculated using the Ballistic Galvanometer were compared with the corresponding micro-ammeter readings

or

The current measured by the Ballistic Galvanometer (BG) is found to be in close agreement with the reading of the micro-ammeter